

TABULATION:-

S.No.	X1		X2		X1		X1		M.F = 2		M.F = 4		efficiency (%)	Voltage Regulation (%)
	I ₁	V ₁	I ₂	V ₂	I ₁	V ₁	I ₂	V ₂	Power input W _{input} x1	Power output W _{output} x2	Power input W _{input} x1	Power output W _{output} x2		
1	0	116 x 2 = 232	0	230	0	230	0	230	0	0	0	0	0	0
2	0.5	112 x 2 = 224	0.5	226	0.5	226	0.5	226	480	360	480	360	75	1.75
3	1.5	112 x 2 = 224	1.5	222	1.5	222	1.5	222	180	750	180	750	85	3.4

Formulae:-

Output power = $W_2 \times M.F$

Input power = $W_1 \times M.F$

$\eta\% = \frac{\text{Output power}}{\text{Input power}} \times 100\%$

$\text{Reg}\% = \frac{V_{NL} - V_{FL}(\text{secondary})}{V_{NL}} \times 100\%$

Name Plate details

Rated voltage = 230 V 115 V

current = 5A 10A

Power = 1kVA 1kVA

efficiency = $1. \frac{0}{0} \times 100 = 0$

$2. \frac{360 \times 100}{480} = 75\%$

$3. \frac{750 \times 100}{880} = 85\%$

Regulation

$1. \frac{230 - 0}{230} \times 100 = 0$

$2. \frac{230 - 226}{230} \times 100 = 1.74\%$

$3. \frac{230 - 222}{230} \times 100 = 3.48\%$

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* Note the readings of primary

voltage (V_1), primary current (I_1), secondary voltage (V_2), secondary current (I_2), power consumed by the load (W_2)

* Calculate the voltage regulation

voltage regulation (%) = $\frac{V_2 - V_1}{V_1} \times 100$

* Calculate efficiency

$\eta = \frac{\text{output}}{\text{input}} \times 100$

* Analyze the results

Calculation:-

$\eta = \frac{P_{out} \times 100}{P_{in}}$

voltage regulation (%) = $\frac{V_1 - V_2}{V_2} \times 100$

1. $\eta = \frac{0}{120} \times 100 = 0$

1. $\% = \frac{230 - 230}{230} \times 100 = 0$

2. $\eta = \frac{360}{480} \times 100 = 75\%$

2. $\% = \frac{230 - 226}{230} \times 100 = 1.74\%$

3. $\eta = \frac{750}{880} \times 100 = 85\%$

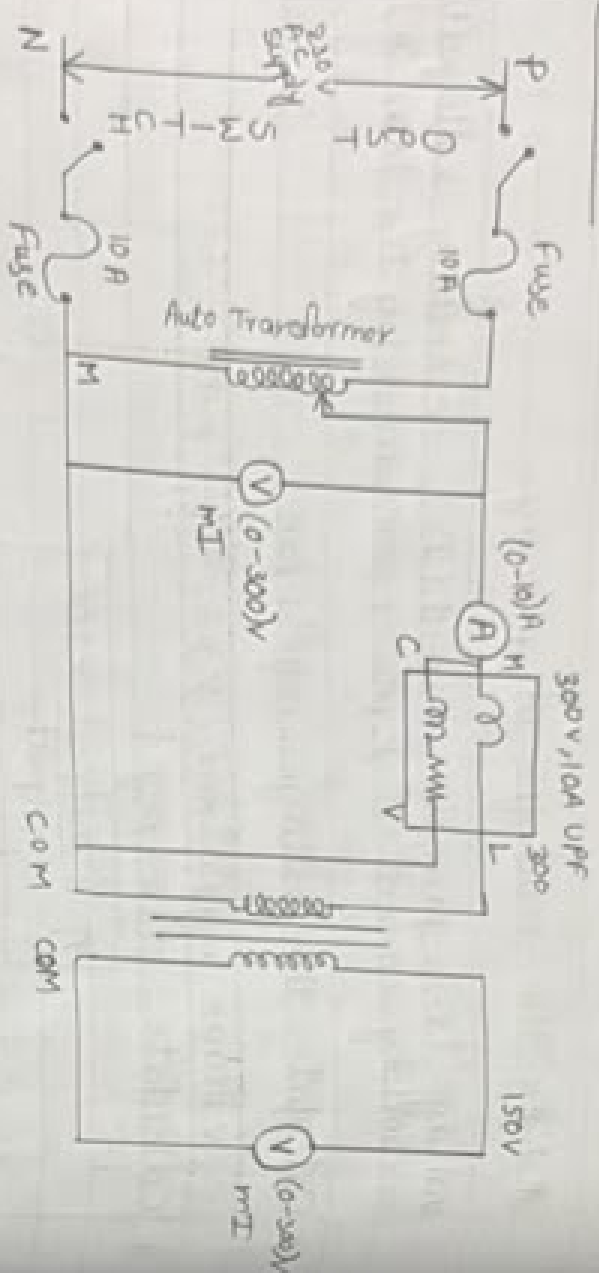
3. $\% = \frac{230 - 222}{230} \times 100 = 3.48\%$

4. $\eta = \frac{320}{420} \times 100 = 76.1\%$

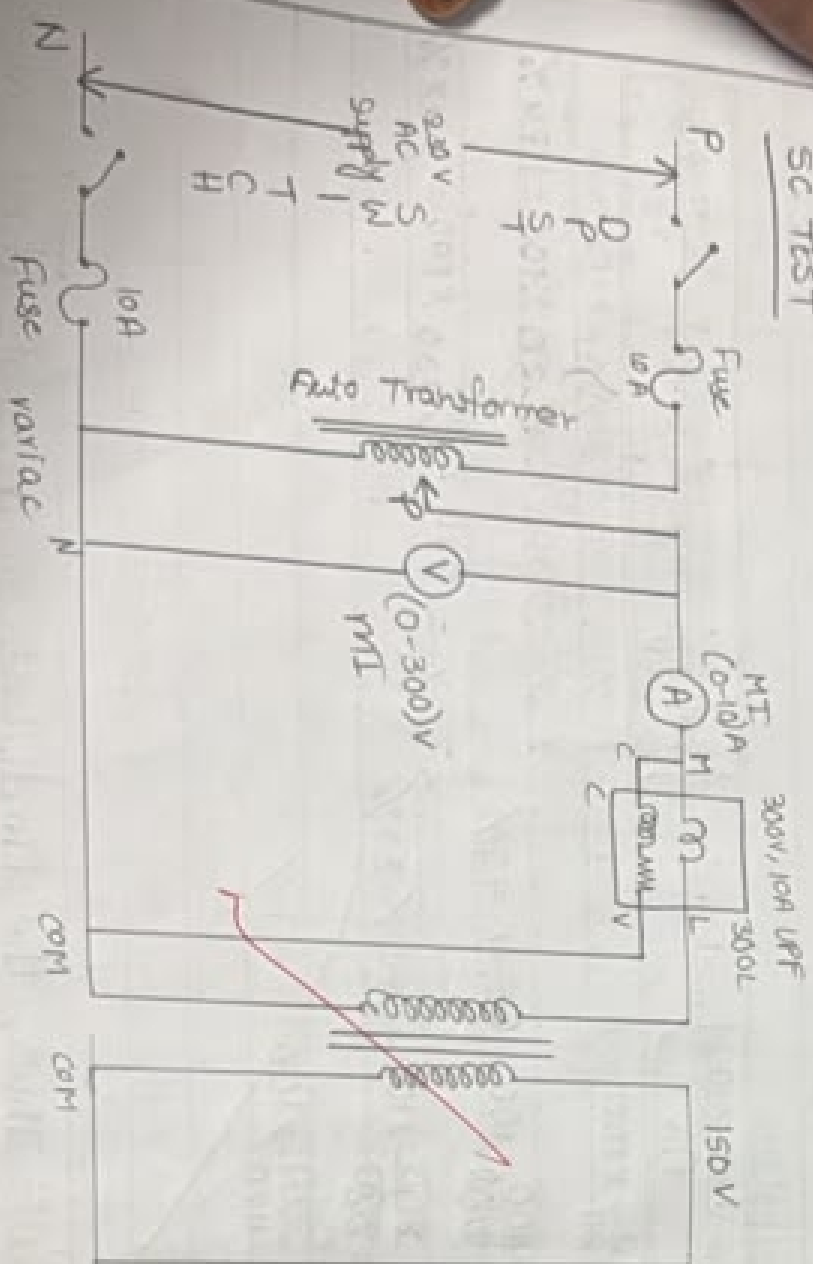
Result:- Thus, the load test on a single phase transformer is successful.

Teacher's Signature _____

OC TEST



SC TEST



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EXP. NO: 09

DC AND SC TEST ON SINGLE PHASE TRANSFORMER

Aim: To conduct load test on single phase transformer to find efficiency and percentage regulation.

Apparatus Required:

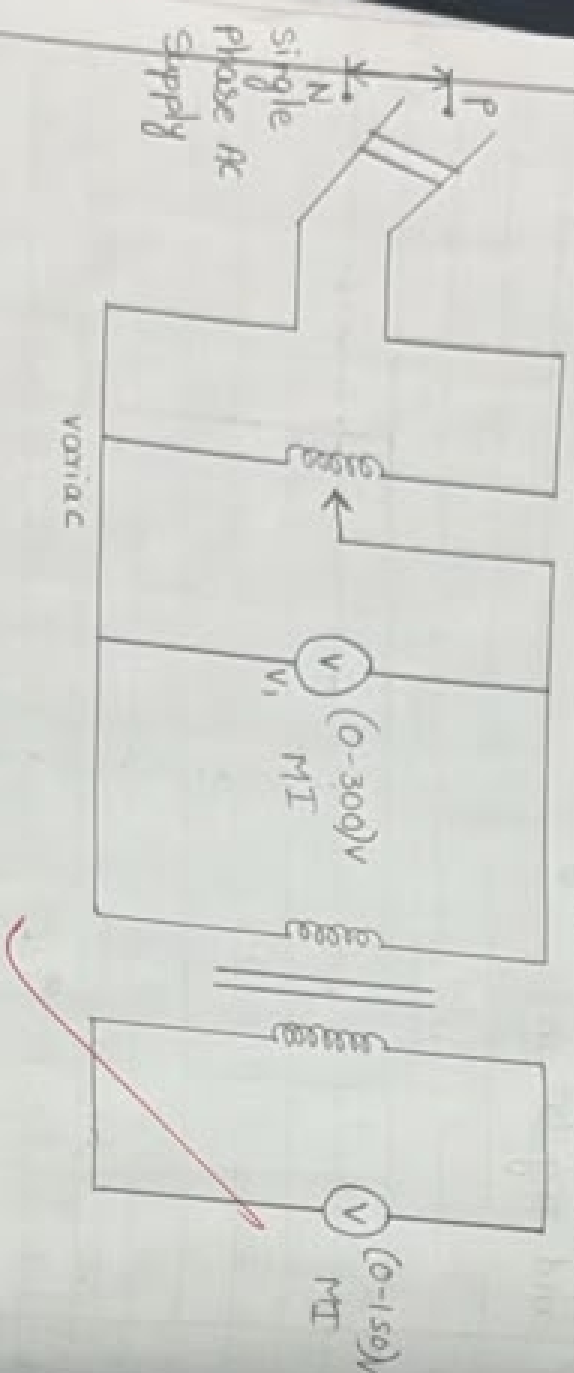
S.No.	Apparatus	Range	Type	Quantity
1.	Ammeter	(0-10) A	MI	1
2.	Voltmeter	(0-150) V	MI	1
3.	Wattmeter	(0-300) W	MI	1
4.	Auto transformer	(0-250) V	UPF	1
5.	Resistive load	5 kW, 230 V	—	1
6.	Connecting wires	2.5 sq. mm	upper	few

Procedure:

1. Connect the primary winding to variable AC supply, attach a ammeter, voltmeter and wattmeter.
2. Run the transformer with no load connected. measure the no-load secondary voltage.
3. Gradually increase the load on the secondary side.
4. Efficiency (η) = $\frac{\text{Output power}}{\text{input power}} \times 100$

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CIRCUIT DIAGRAM:



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EXP. No: 10 CALCULATION OF SECONDARY TURNS AND CURRENT IN A TRANSFORMER

Aim: To evaluate secondary turns and current in a transformer and to find voltage regulation, efficiency and overall behaviour of transformer when it is delivering power to a load.

Apparatus Required:-

S.No.	Apparatus	Range	Quantity
1.	Single phase transformer	1 ϕ , (0-250)V	1
2.	Voltmeter	(0-150V), (0-300V) MT Type	1 each
3.	Ammeter	(0-10A), (0-5A) MT Type	1 each
4.	Wattmeter	(300V, 5A), (150V, 5A) UF	1 each
5.	Supply source (AC power source)		
6.	Variable resistive load or load bank	5KW, 230V	1
7.	Connecting wires	as required	

Procedure:-

① A transformer operates on the principle of electromagnetic induction and consists of primary winding, secondary winding and turn ratio.

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② Before performing calculations, gather the required parameters like V_p , N_p , I_p , V_s .

③ Calculate secondary current (I_s)

$$I_s = (N_p/N_s) \times I_p$$

④ Analyze the result.

Calculations:

$$I_2 = 17/0.5 = 34$$

$$I_2 = 13/0.5 = 26$$

$$I_2 = 10/0.5 = 20$$

$$I_2 = 8/0.5 = 16$$

Result: By following these steps, you can effectively calculate the secondary turns and current in a transformer. This procedure can be adapted for any set of input values, making it a versatile method for transformer analysis. Thus the calculation of secondary turns and current in a transformer.

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TABULATION:-

S.No.	Primary Voltage (V ₁)	Secondary Voltage (V ₂)	Primary Current in amp	Turn ratio $K = V_1/V_2$	Secondary Current $I_2 = I_1/K$
1	60	30	17	0.5	34
2	80	40	13	0.5	26
3	100	50	10	0.5	20
4	120	60	8	0.5	16

TABULATION:-

S. No	Load Current	Secondary Voltage	Primary Voltage	Primary Current	Power Input	Power Output	Efficiency	Voltage Regulation
1	0	230	230	0	0	0	0	0
2	0.3	220	228	1	212	100	94.3	4.5%
3	0.8	216	228	1.8	500	480	96	6.4%
4	2.1	212	224	2.6	740	720	97.2	8.4%

to the min voltage position and DFST is opened.
 The radius of the brake drum is measured.

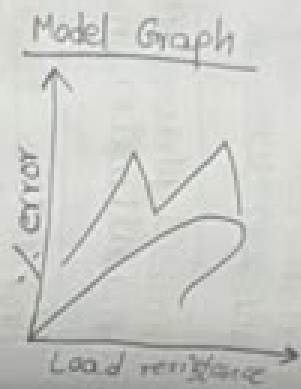
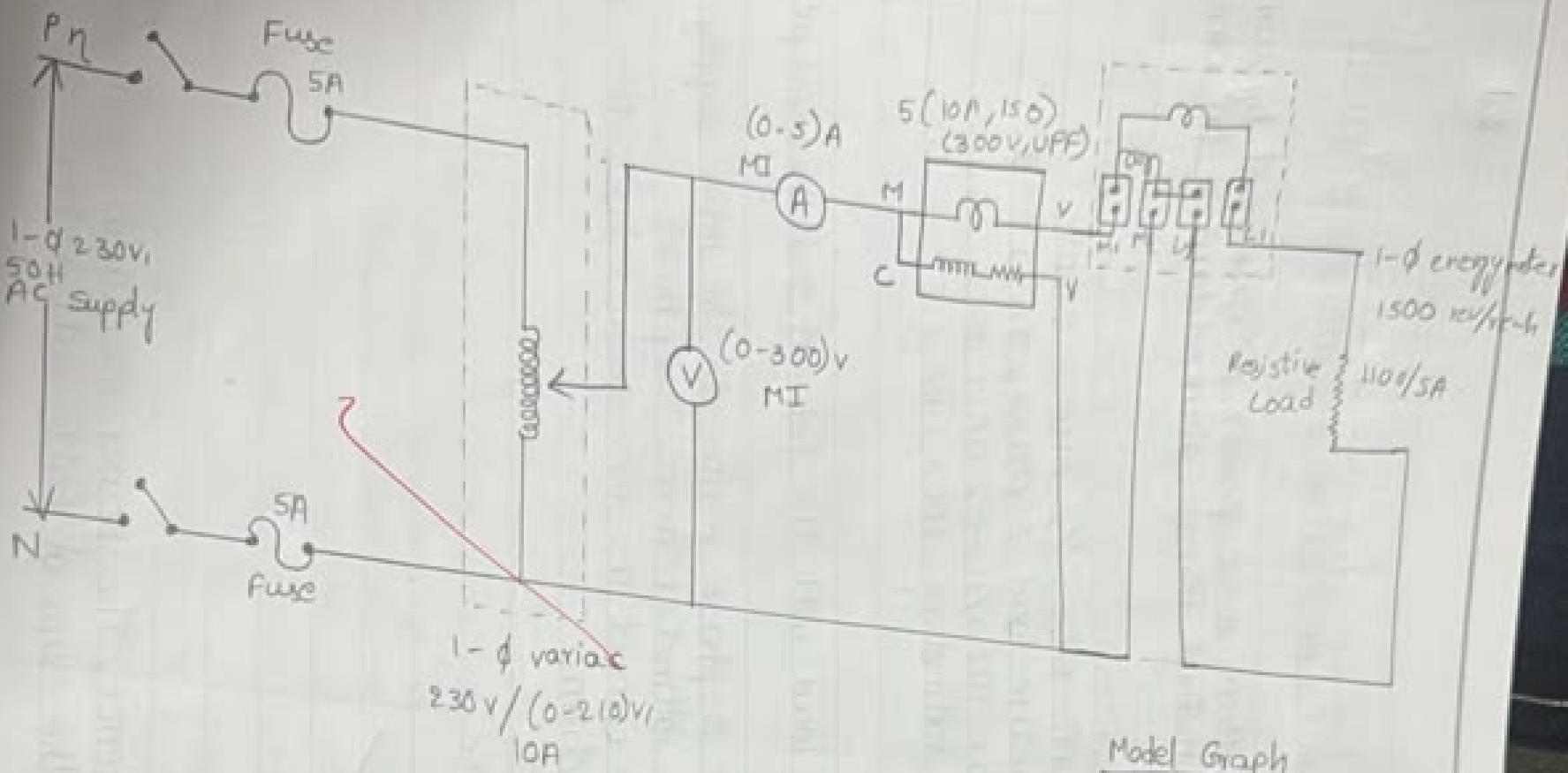
Formulae:-

- * Circumference of brake drum = $2\pi R$
- * input power w (watts)
- * Torque $(T) = 9.8 / \cdot R + (S_1 - S_2)(N \cdot m)$
- * Output power = $2\pi NT / 60$ (watts)
- * % efficiency $(\eta) = \text{Output Power} / \text{input power} \times 100$
- * Power factor, $\cos \phi = W_1 / VT$
- * % slip, $S = \frac{N_s - N}{N_s} \times 100$ [$N_s = 120f / p$]

Result:- Thus, the load test was conducted on 1 ϕ induction motor & the performance characteristics were drawn.

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CIRCUIT DIAGRAM:-



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EXP. NO: 13

CALCULATE THE ENERGY CONSUMPTION USING THE ENERGY METER

Aim: To calibrate and test the given single phase energy meter by circuit loading.

Apparatus Required:

S.No.	Name	Type	Range	Quantity
1.	Single phase energy meter	Induction	1500 Rev / kWh	1
2.	Wattmeter	VPF	300V/5A	1
3.	Voltmeter	MI	(0-300)V	1
4.	Ammeter	MI	(0-5)A	1
5.	Single phase variac	1-φ	230V/(0-240)	1
6.	Rheostat	kW	V, 10A	1
7.	stop watch	digital	110-2/5A	1

Procedure:

1. Connect the circuit as per circuit diagram.
2. Keep the single phase variac at given volt position.
3. Now switch on the power supply.
4. Gradually vary the variac to apply the rated voltage (230V).
5. For different values of load, note down the readings of the ammeter, voltmeter, wattmeter, and time taken for 10 revolutions of the disc.
6. Gradually vary the variac to minimum on zero volt position.

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TABULATION:-

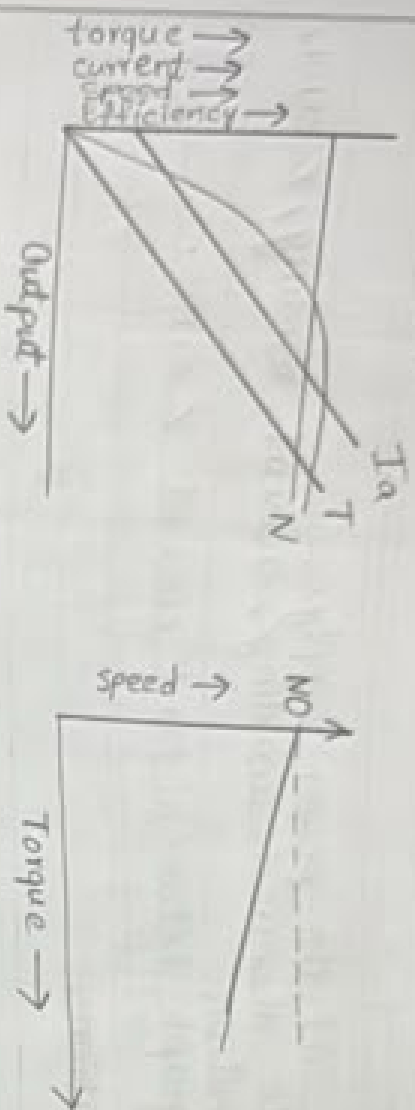
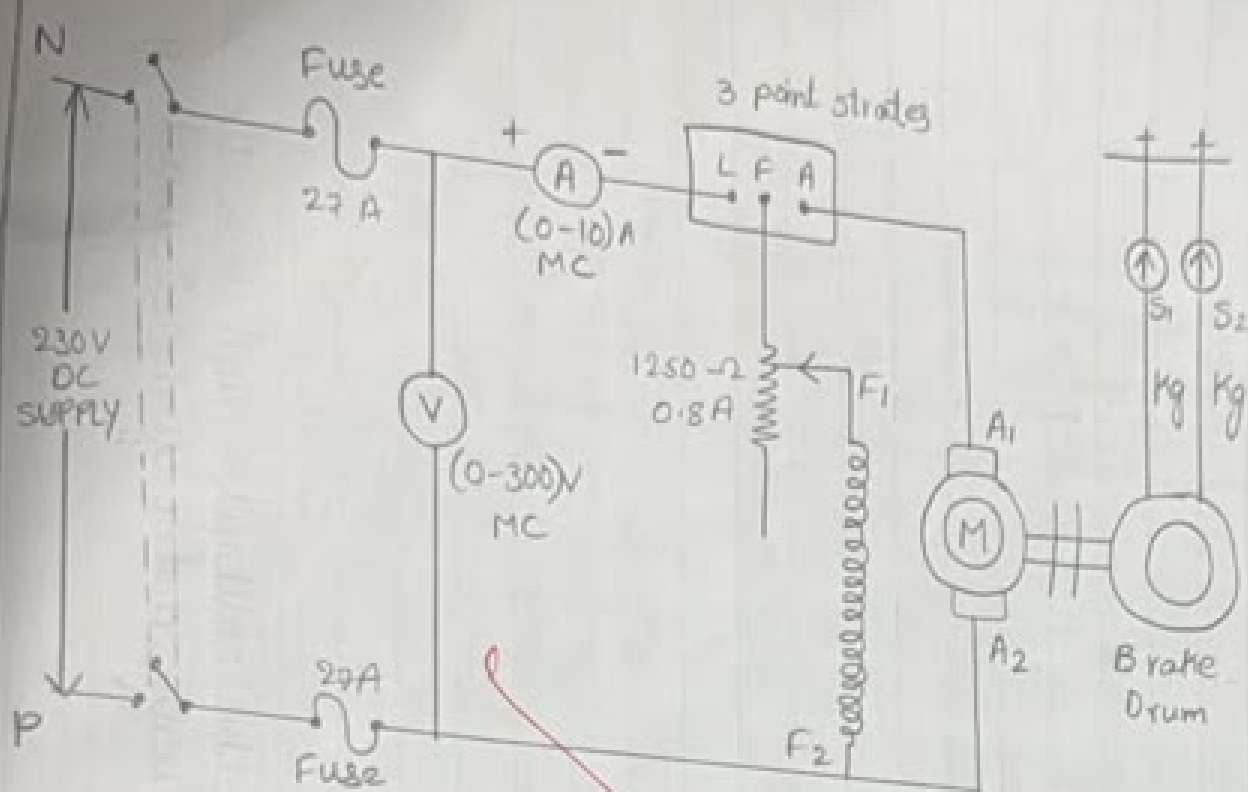
S.No.	Voltmeter (volts)	Ammeter (Amps)	No. of revolution	Time for 10 rev (sec)	Energy	Measured Energy
1	230	1	10	8.6 sec	5.433×10^{-3}	6.66×10^{-3}
2	225	3	10	3.8 sec	7.125×10^{-3}	6.66×10^{-3}

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- Switch off the power supply.
 - Calculate observed reading, actual reading, % error, % correction.
 - Draw graph between load current % error.

Result:- Hence, the calibrate and test the single phase energy meter by direct loading is done.

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CIRCUIT DIAGRAM:-



MODEL GRAPH:-

Exp. No:- 14 LOAD TEST ON DC SHUNT MOTOR

Aim:- To conduct the brake load test on DC shunt motor, determine its performance characteristics.

APPARATUS REQUIRED:-				
S.No.	Apparatus	Range	Type	Quantity
1.	Ammeter	(0-30)A	digital	1
2.	Voltmeter	(0-300)V	digital	1
3.	Rheostat	310-1.7A	digital	1
4.	RPM meter	(0-9999)rpm	digital	1
5.	Connecting wires	-	-	As required
6.	Tachometer	-	-	-

Procedure:-

1. Make the connections as shown in circuit diagram.
2. Keeping the field rheostat (Rf) at the minimum position, switch on the supply and start the motor.
3. Adjust the speed of the motor on no load to its rated value by means of the field rheostat. Do not disturb the position of the rheostat through the test.
4. Put on the load by tightening the screw's of the spring balances. Note down the voltage and current at different loads until full load current obtained.

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TABULATION:-

S.No.	Voltage	Current	Speed IN rpm	Spring Balance	Torque (NM) (T)	Output Power (W) $W = \frac{2\pi NT}{60}$	INPUT POWER (W) VI	Efficiency η IN %
1.	220	15.6	0	0	0	0	352	0%
2.	220	4	0.6	2.2	1.412	212.15	880	23.2%
3.	220	6	0.8	5.2	3.884	569.135	1320	43%
4.	220	8	1	8	6.1803	893.9	1760	50%
5.	218	10	1.2	10.2	7.946	1146.05	2180	52.5%

Result:- Thus, the performance characteristics of DC shunt motor was obtained by conducting brake test.

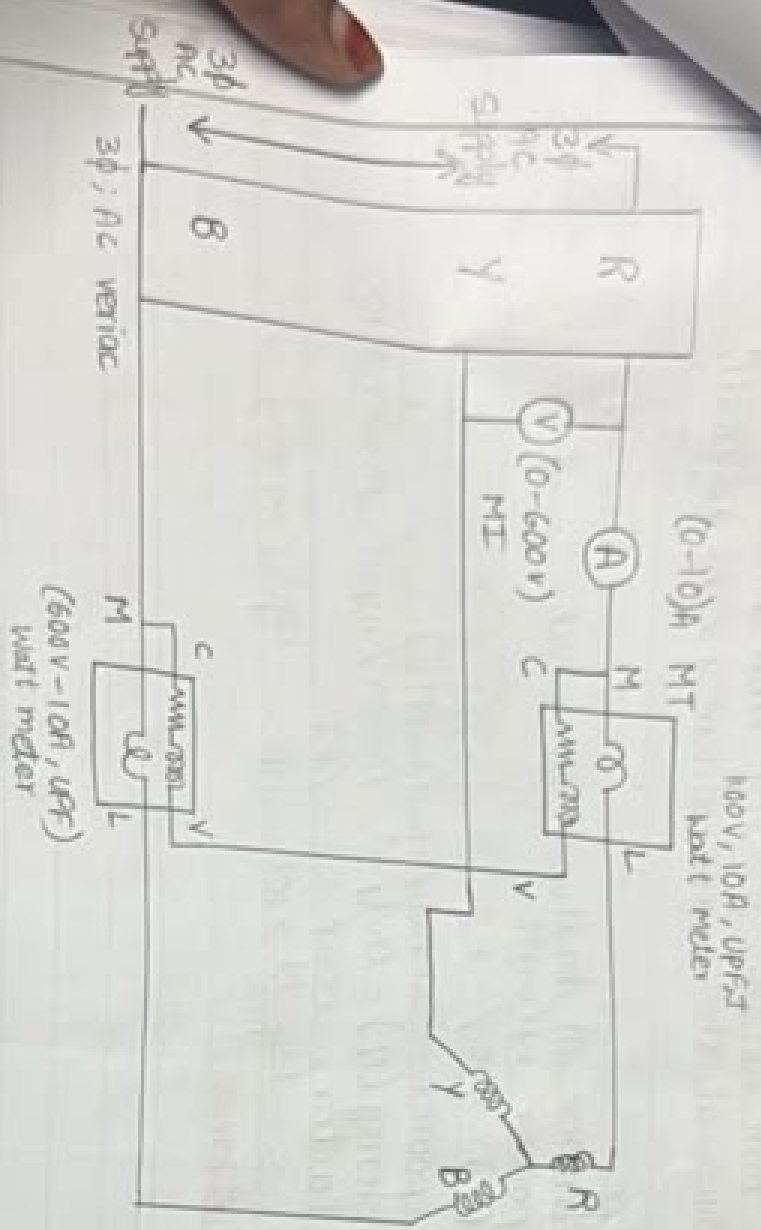
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CIRCUIT DIAGRAM:-



TRIBUTION:-

S.No.	V _L	I _L	W ₁	W ₂	P = W ₁ + W ₂	P _T = I ₃ V _L
1	175	2	320	280	600	606.2
2	185	2.1	335	347	682	672.9
3	195	2.2	370	390	760	753.2
4	205	2.3	410	430	846	830.9

EXP. NO:-12 MEASUREMENT OF POWER BY TWO WATTMETER METHOD

Aim:- 3 phase, three wire power measurement by using 2 wattmeter method for a balanced load in star connection.

Apparatus Required:-

1. Wattmeters 4PF 600V, 1/2 AMPs - 2 Nos.
2. Balanced resistive load 3 phase, 3A-1 No's
3. 3-Phase variac 415 V/470, 4A 1 No's
4. Digital AC Voltmeters 600V-100

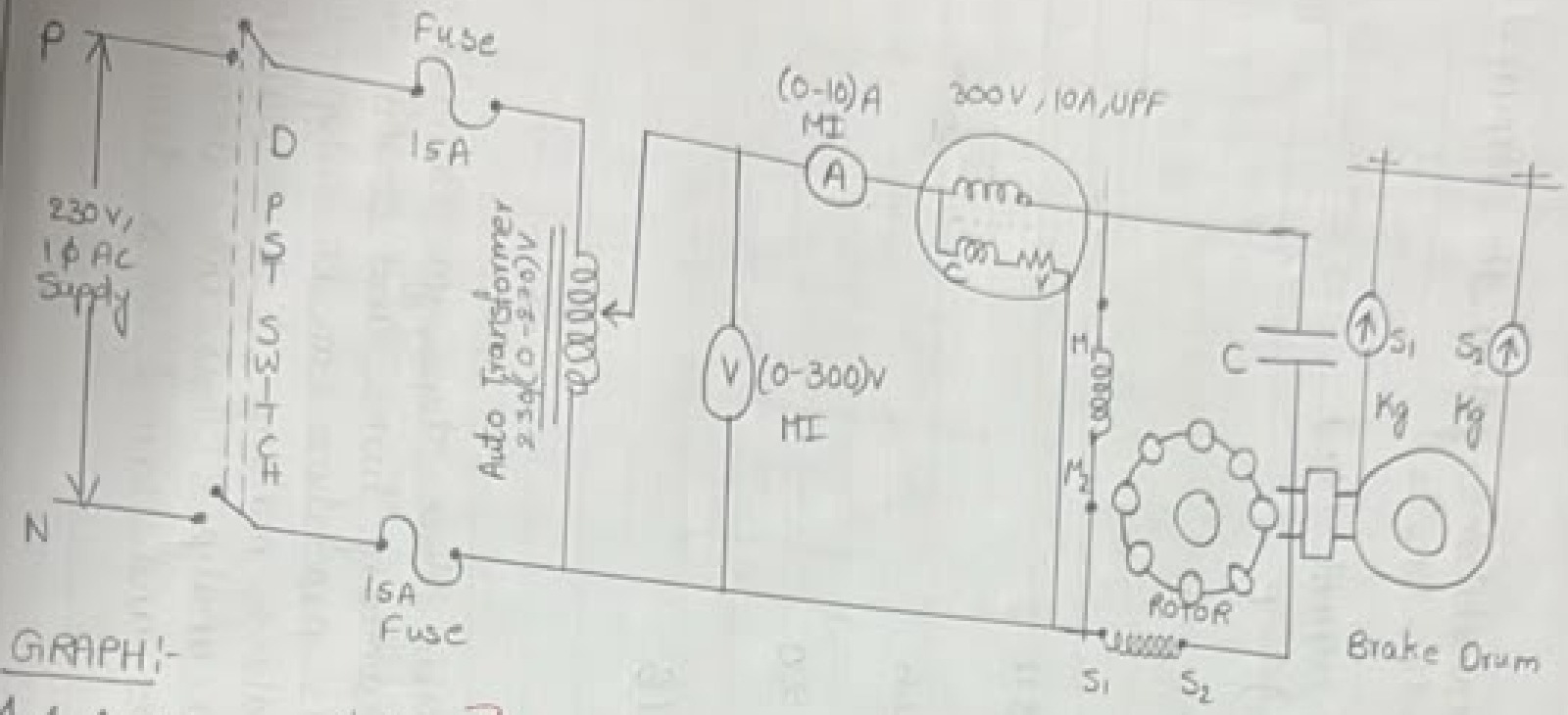
Procedure:-

1. Connect the load with the help of switches and patch chords.
2. Connect the 3-phase variac to the main supply.
3. Connect the wattmeter across 2 phases.
4. Connect the circuit as shown below either in star connection.

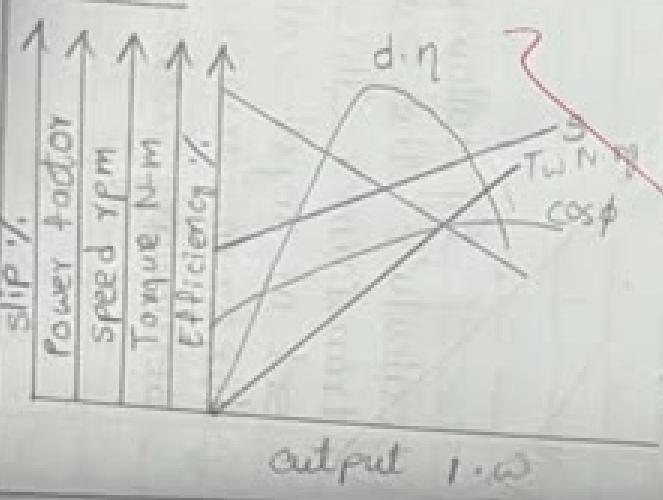
Result:- Hence, the power measured for a balanced load is the sum of both watt meter.

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CIRCUIT DIAGRAM:-



GRAPH:-



NAME PLATE DETAILS

Rated Voltage : 220V
 Rated Current : 9.5A
 Rated Power : 3HP
 Rated Speed : 1490 RPM

S1, S2 - AUXILIARY WINDING
 M1, M2 - MAIN WINDING

EXP. NO. 11

LOAD TEST ON SINGLE PHASE INDUCTION MOTOR

Aim:- To conduct load test on the given single phase induction motor and to plot its performance characteristics.
(i) Electrical characteristics: speed, torque, slip, power factor and efficiency V_s output power.

Apparatus Required:-

S.No.	Apparatus	Specifications	Quantity
1	Voltmeter	(0-300V) M2	1
2	Ammeter	(0-10A) M1	1
3	Wattmeter	(300V, 10A, WPF)	1
4	Tachometer	(0-10000 RPM)	1

Procedure:-

- Connections as per circuit diagram.
- The DPST switch is closed and single phase supply is given.
- Adjust the variac to apply the rated voltage and record no-load readings of speed, spring balance and wattmeters. If a wattmeter shows a -ve reading interchange its pressure and current coil terminals, then restart and rerecord.
- The procedure is repeated till rated current of the machine.
- The motor is unloaded, the auto transformer is brought back to zero.

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OC:-

TEST	Voltage (V)	Current (A)	Power (W)
OC test	$V_o = 230V$	$I_o = 0.5A$	$W_o = 60W$
SC test	$V_{sc} = 20V$	$I_{sc} = 4.2A$	$W_{sc} = 96W$

SC:-

S.No	Volt	Amps	Speed (rpm)	Spring balance			Input	Output	Reg. %	N = $\frac{E}{I_a}$
				S1	S2	S2-S1				
1	230	3.2	1490	300	0	0	7.36	0	0.44%	0
2	230	3.5	1480	540	0.0	1	0.2	0.196	80.5	10.54
3	228	4	1460	760	1.6	2	0.4	0.372	91.2	57.11
4	226	4.5	1450	900	2	2.8	0.8	0.684	101	111.10

⑤ Voltage regulation calculation

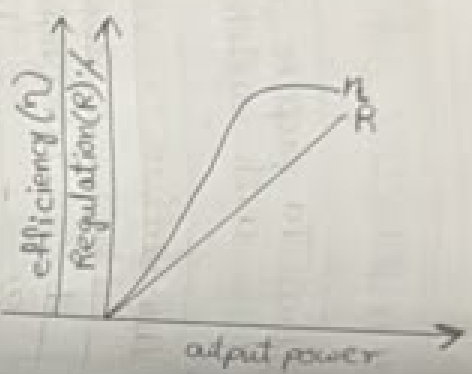
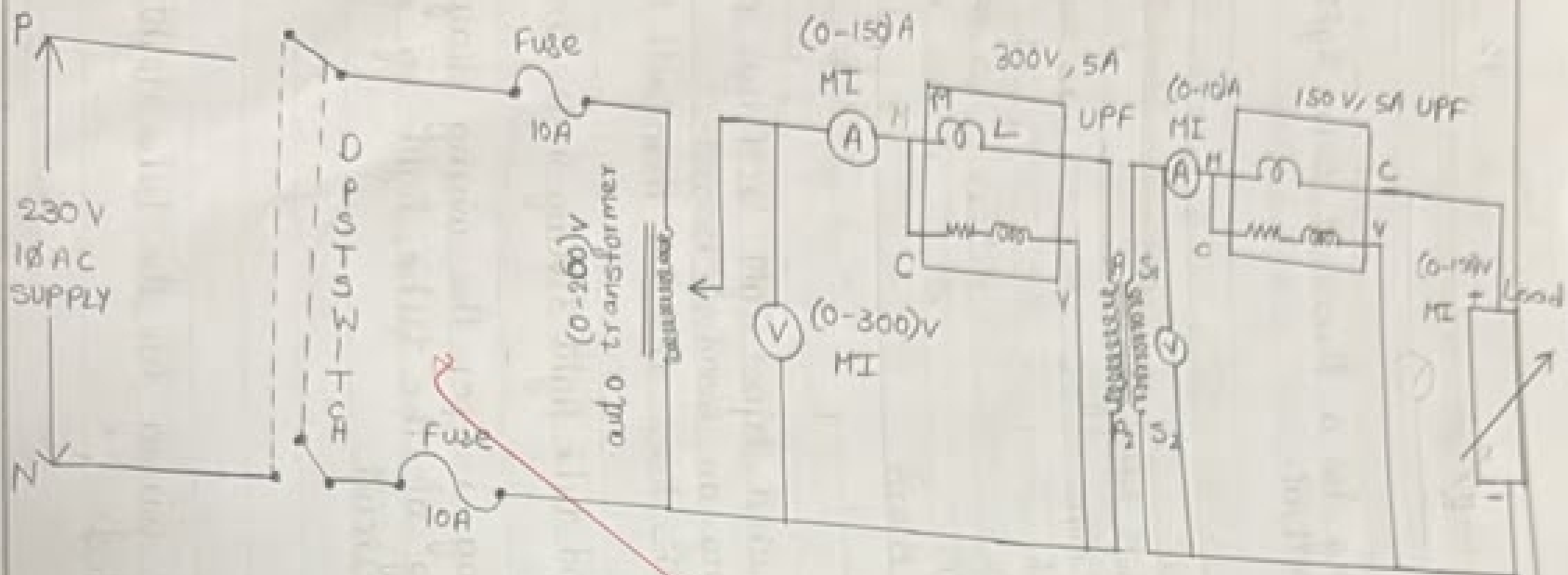
$$\% \text{ Regulation} = \frac{V_2 - V_1}{V_1} \times 100$$

⑥ Tabulate the readings and calculate efficiency and regulation.

Result:- Thus, the OC & SC test on single phase transformer is conducted.

Teacher's Signature _____

CIRCUIT DIAGRAM



Exp. No. _____

Date _____ Page No. 13

EXP. NO. 08 LOAD TEST ON SINGLE PHASE TRANSFORMER

A load test on a single phase transformer is conducted to evaluate the performance of transformers under full load capacitor container.

APPARATUS REQUIRED:		
S.No.	APPARATUS	RANGE
1.	Single phase transformer	1Ø (0-200)V
2.	Voltmeter	(0-300)V (0-150)V MI type
3.	Ammeter	(0-10)A (0-300) mA MI type
4.	Watt meter	(300V, 5A) (150V, 5A) UPF
5.	Supply Source	
6.	Variable resistive load or load bank	5kW, 230V
7.	Connecting wires	as required

Procedure:

1. Connect the circuit as per required.
2. Supply the rated voltage to primary winding using a variac (or) auto transformer.
- * Adjust the load.

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